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**BIT 2324 GEOGRAPHICAL INFORMATION
SYSTEMS**

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke



LESSON 3

Projections & Coordinate Systems

Learning outcomes

By the end of this topic you should be able to;

- *Define coordinate systems*
- *Describe different types of maps*
- *Distinguish vector, raster & image data formats*



3.1. Introduction

A map is a graphical representation of where features are, explicitly and relative to one another. It is composed of different geographic features represented as either points, lines, and/or areas. Each feature is defined both by its location in space and by its characteristics. Quite simply, a map is a model of the real world. The map legend is the key linking the attributes to the geographic features. Attributes, e.g. such as the species for a forest stand, are typically represented graphically by use of different symbology and color.

Maps and mapping are essential components of any and all geographic information systems (GISs). For instance, maps constitute both the input and output of a GIS. Hence a GIS utilizes many concepts and themes from cartography, the formal study



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of maps and mapping. In order to become proficient with GISs, there is need to learn more about cartography, maps, and mapping.

3.2. Map Scale, Coordinate Systems, and Map Projections

The central purpose of a map is to provide relevant and useful information to the map user. In order for a map to be of value, it must convey information effectively and efficiently. Mapping conventions facilitate the delivery of information in such a manner by recognizing and managing the expectations of map users. Mapping or cartographic conventions refer to the accepted rules, norms, and practices behind the making of maps. One of the most recognized mapping conventions is that “north is up” on



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most maps. Though this may not always be the case, many map users expect north to be oriented or to coincide with the top edge of a map or viewing device like a computer monitor.

All map users and map viewers have certain expectations about what is contained on a map. Such expectations are formed and learned from previous experience by working with maps. It is important to note that such expectations also change with increased exposure to maps. Understanding and meeting the expectations of map viewers is a challenging but necessary task because such expectations provide a starting point for the creation of any map.

Several other formal and informal mapping conventions and characteristics, many of which are taken for granted, can be identified. Among the most important cartographic considerations



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are map scale, coordinate systems, and map projections. Map scale is concerned with reducing geographical features of interest to manageable proportions, coordinate systems help us define the positions of features on the surface of the earth, and map projections are concerned with moving from the three-dimensional world to the two dimensions of a flat map or display.

3.2.1. Map Scale

One of the challenges behind mapping the world and its resident features, patterns, and processes is reducing it to a manageable size. All maps reduce or shrink the world and its geographic features of interest by some factor. Map scale refers to the factor of reduction of the world so that it fits on a map.

Map scale can be represented by text, a graphic, or some



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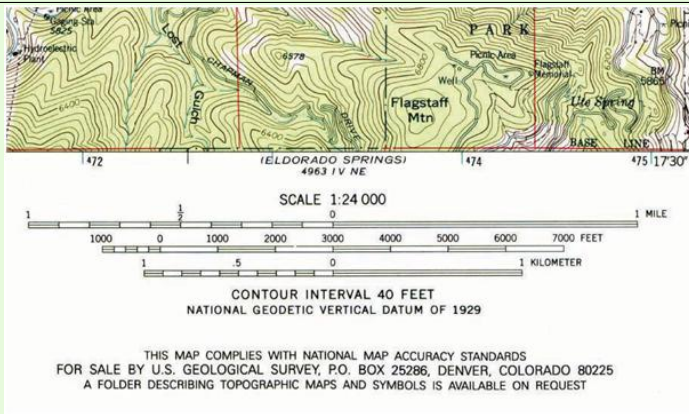
combination of the two. For example, it is common to see “one inch represents one kilometer” or something similar written on a map to give map users an idea of the scale of the map. Map scale can also be portrayed graphically with what is called a scale bar. Scale bars are usually used on reference maps and allow map users to approximate distances between locations and features on a map, as well as to get an overall idea of the scale of the map.

Map Scale from a United States Geological Survey (USGS)
Topographic Map





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The representative fraction (RF) describes scale as a simple ratio. The numerator, which is always set to one (i.e., 1), denotes map distance and the denominator denotes ground or “real-world” distance. One of the benefits of using a representa-



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tive fraction to describe scale is that it is unit neutral. Any unit of measure can be used to interpret the map scale. Consider a map with an RF of 1:10,000. This means that one unit on the map represents 10,000 units on the ground. Such units could be inches, centimeters, or even pencil lengths; it really does not matter.

Map scales can also be described as either “small” or “large.” Such descriptions are usually made in reference to representative fractions and the amount of detail represented on a map. For instance, a map with an RF of 1:1,000 is considered a large-scale map when compared to a map with an RF of 1:1,000,000 (i.e., $1:1,000 > 1:1,000,000$). Furthermore, while the large-scale map shows more detail and less area, the small-scale map shows more area but less detail. Clearly, determining the thresholds



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for small- or large-scale maps is largely a judgment call.

All maps possess a scale, whether it is formally expressed or not. Though some say that online maps and GISs are “scaleless” because we can zoom in and out at will, it is probably more accurate to say that GISs and related mapping technology are multiscalar. Understanding map scale and its overall impact on how the earth and its features are represented is a critical part of both map making and GISs.

3.2.2. Coordinate Systems

Just as all maps have a map scale, all maps have locations, too. Coordinate systems are frameworks that are used to define unique positions. For instance, in geometry we use x (horizontal) and y (vertical) coordinates to define points on a two-



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dimensional plane. The coordinate system that is most commonly used to define locations on the three-dimensional earth is called the geographic coordinate system (GCS), and it is based on a sphere or spheroid. A spheroid (a.k.a. ellipsoid is simply a sphere that is slightly wider than it is tall and approximates more closely the true shape of the earth. Spheres are commonly used as models of the earth for simplicity.

The unit of measure in the GCS is degrees, and locations are defined by their respective latitude and longitude within the GCS. Latitude is measured relative to the equator at zero degrees, with maxima of either ninety degrees north at the North Pole or ninety degrees south at the South Pole. Longitude is measured relative to the prime meridian at zero degrees, with maxima of 180 degrees west or 180 degrees east.

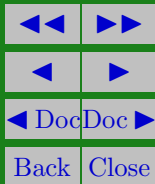


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Note that latitude and longitude can be expressed in degrees-minutes-seconds (DMS) or in decimal degrees (DD). When using decimal degrees, latitudes above the equator and longitudes east of the prime meridian are positive, and latitudes below the equator and longitudes west of the prime meridian are negative (see the following table for examples).

Converting from DMS to DD is a relatively straightforward exercise. For example, since there are sixty minutes in one degree, we can convert $118^{\circ} 15$ minutes to 118.25 ($118 + 15/60$). Note that an online search of the term “coordinate conversion” will return several coordinate conversion tools.

When we want to map things like mountains, rivers, streets, and buildings, we need to define how the lines of latitude and longitude will be oriented and positioned on the sphere. A da-





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tum serves this purpose and specifies exactly the orientation and origins of the lines of latitude and longitude relative to the center of the earth or spheroid.

Depending on the need, situation, and location, there are several datums to choose from. For instance, local datums try to match closely the spheroid to the earth's surface in a local area and return accurate local coordinates. A common local datum used in the United States is called NAD83 (i.e., North American Datum of 1983). For locations in the United States and Canada, NAD83 returns relatively accurate positions, but positional accuracy deteriorates when outside of North America.

The global WGS84 datum (i.e., World Geodetic System of 1984) uses the center of the earth as the origin of the GCS and is used for defining locations across the globe. Because the datum



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uses the center of the earth as its origin, locational measurements tend to be more consistent regardless where they are obtained on the earth, though they may be less accurate than those returned by a local datum. Note that switching between datums will alter the coordinates (i.e., latitude and longitude) for all locations of interest.

For geographic data, often referred to as spatial data, features are usually referenced in a coordinate system that models a location on the earth's surface. The coordinate system may be of a variety of types. For natural resource applications the most common are geographic coordinates such as latitude and longitude, e.g. $56^{\circ}27'40''$ and $116^{\circ}11'25''$. These are usually referred by degrees, minutes, and seconds. Geographic coordinates can also be identified as decimal degrees, e.g. 54.65° .



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Geographic data is distinguished from attribute data in that it is referenced spatially by a coordinate system, e.g. it has a spatial extent. Natural resource applications commonly use a Legal Survey system.

- **Entity Representations**

We need symbolize spatial features in order to be able to associate attribute information. We can classify different features into different dimensions. Each classification of dimension is a conceptual classification.

- **Points** - “0” dimensionality. No length or Width.: Each point is Discrete in that it can only occupy a given point in space at any given time.
- **Lines** - “1” dimensional. Length, but No Width: Must



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have a beginning and an ending point.

- **Polygons** - “2” dimensional. Length and Width: By adding Width, we can describe a feature as having an area.
- **Surfaces** - “3” dimensional. Length, Width, and Height: Surfaces have infinite number of values (e.g. Elevation). We say that this type of data is Continuous.

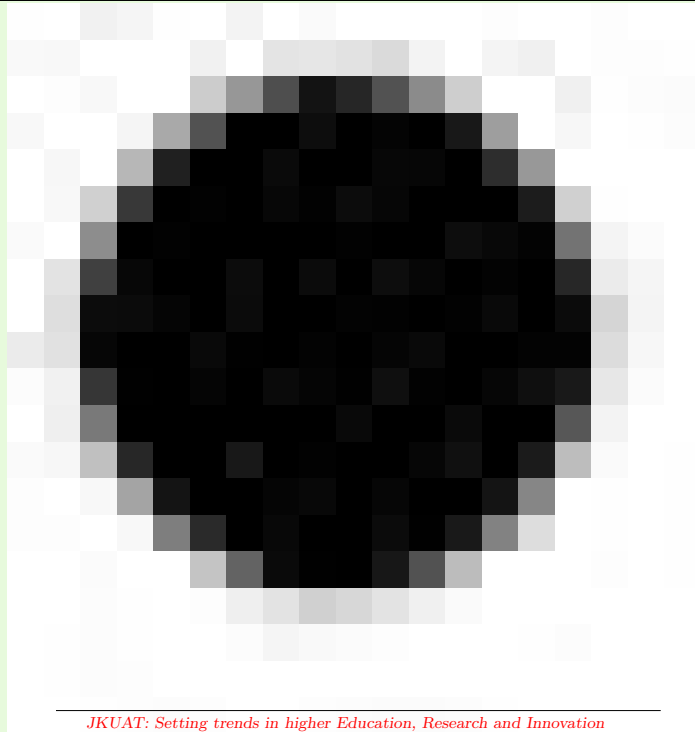
When thinking of spatial elements, we must consider Spatial Scale. Depending on scale, we may want to represent a river as a line or a polygon.

We typically represent objects in space as three distinct spatial elements:

1. Points - simplest element



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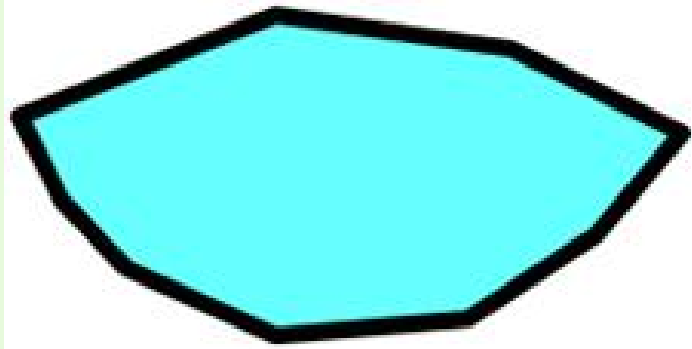


2. Lines (arcs) - set of connected points



3. Polygons - set of connected lines





3.2.3. Map Projections

Map projections refer to the methods and procedures that are used to transform the spherical three-dimensional earth into two-dimensional planar surfaces. Specifically, map projections are mathematical formulas that are used to translate latitude and



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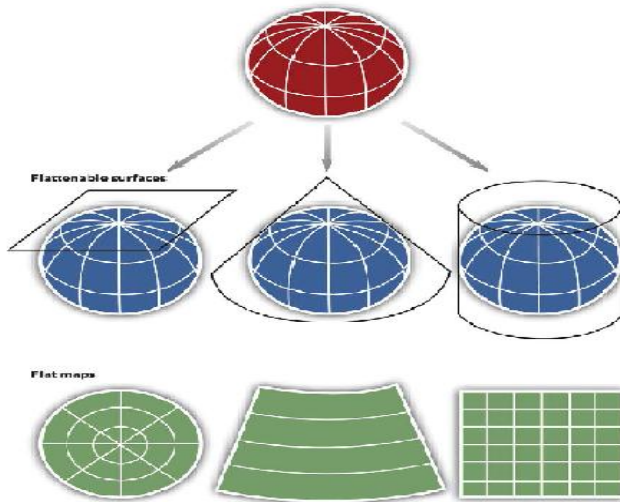
longitude on the surface of the earth to x and y coordinates on a plane. Since there are an infinite number of ways this translation can be performed, there are an infinite number of map projections. The mathematics behind map projections are beyond the scope of this unit. To illustrate the concept of a map projection, imagine that we place a light bulb in the center of a translucent globe. On the globe are outlines of the continents and the lines of longitude and latitude called the graticule. When we turn the light bulb on, the outline of the continents and the graticule will be “projected” as shadows on the wall, ceiling, or any other nearby surface. This is what is meant by map “projection.”

Within the realm of maps and mapping, there are three surfaces used for map projections (i.e., surfaces on which we project the shadows of the graticule). These surfaces are the plane, the



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cylinder, and the cone. Referring again to the previous example of a light bulb in the center of a globe, note that during the projection process, we can situate each surface in any number of ways. For example, surfaces can be tangential to the globe along the equator or poles, they can pass through or intersect the surface, and they can be oriented at any number of angles.

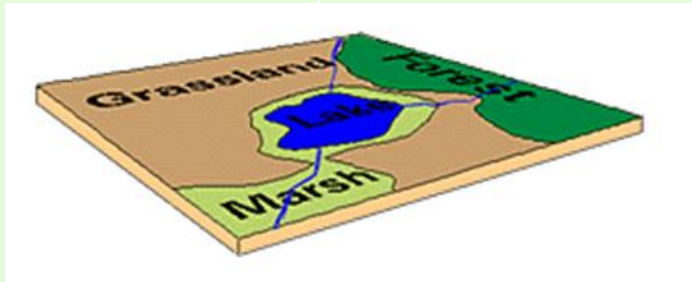




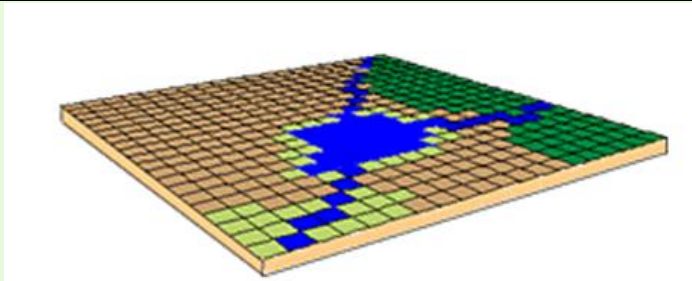
3.3. Spatial Data Models

Traditionally spatial data has been stored and presented in the form of a map. Three basic types of spatial data models have evolved for storing geographic data digitally. These are referred to as:

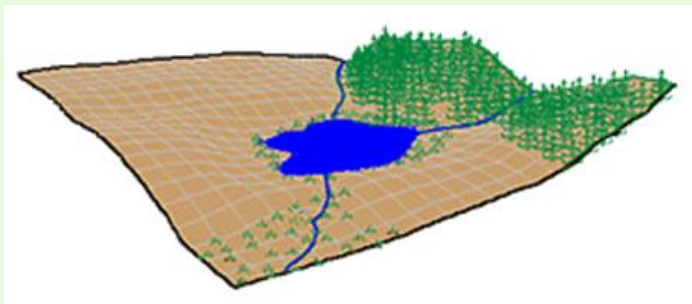
1. Vector;



2. Raster;



3. Image



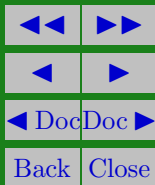


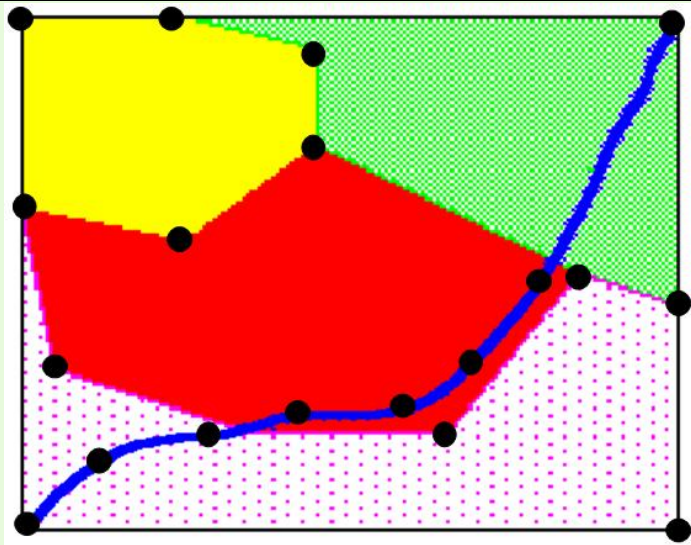
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The below diagrams reflects the two primary spatial data encoding techniques. These are vector and raster. Image data utilizes techniques very similar to raster data, however typically lacks the internal formats required for analysis and modeling of the data. Images reflect pictures or photographs of the landscape.

1. Vector data formats

All spatial data models are approaches for storing the spatial location of geographic features in a database. Vector storage implies the use of vectors (directional lines) to represent a geographic feature. Vector data is characterized by the use of sequential points or vertices to define a linear segment. Each vertex consists of an X coordinate and a Y coordinate.





Vector lines are often referred to as arcs and consist of a string of vertices terminated by a node. A node is defined



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as a vertex that starts or ends an arc segment. Point features are defined by one coordinate pair, a vertex. Polygonal features are defined by a set of closed coordinate pairs. In vector representation, the storage of the vertices for each feature is important, as well as the connectivity between features, e.g. the sharing of common vertices where features connect.

Several different vector data models exist, however only two are commonly used in GIS data storage. The topologic data structure is often referred to as an intelligent data structure because spatial relationships between geographic features are easily derived when using them. Primarily for this reason the topologic model is the dominant vector data structure currently used in GIS technology. Many of



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the complex data analysis functions cannot effectively be undertaken without a topologic vector data structure.

The secondary vector data structure that is common among GIS software is the computer aided drafting (CAD) data structure. This structure consists of listing elements, not features, defined by strings of vertices, to define geographic features, e.g. points, lines, or areas. There is considerable redundancy with this data model since the boundary segment between two polygons can be stored twice, once for each feature. The CAD structure emerged from the development of computer graphics systems without specific considerations of processing geographic features. Accordingly, since features, e.g. polygons, are self-contained and independent, questions about the adjacency of features can



be difficult to answer. The CAD vector model lacks the definition of spatial relationships between features that is defined by the topologic data model.

- **Vector Advantages**

- Accurate positional information that is best for storing discrete thematic features (e.g., roads, shorelines, sea-bed features).
- Compact data storage requirements
- Can associate unlimited numbers of attributes with specific features

2. Raster data formats

Raster data models incorporate the use of a grid-cell data structure where the geographic area is divided into cells identified by row and column. This data structure is com-



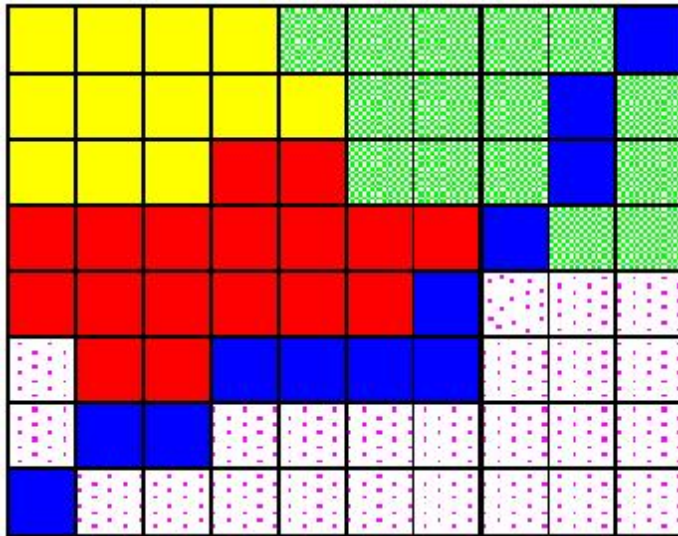
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monly called raster. While the term raster implies a regularly spaced grid other tessellated data structures do exist in grid based GIS systems. In particular, the quadtree data structure has found some acceptance as an alternative raster data model. The size of cells in a tessellated data structure is selected on the basis of the data accuracy and the resolution needed by the user. There is no explicit coding of geographic coordinates required since that is implicit in the layout of the cells. A raster data structure is in fact a matrix where any coordinate can be quickly calculated if the origin point is known, and the size of the grid cells is known. Since grid-cells can be handled as two-dimensional arrays in computer encoding many analytical operations are easy to program. This makes tessellated



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data structures a popular choice for many GIS software. Topology is not a relevant concept with tessellated structures since adjacency and connectivity are implicit in the location of a particular cell in the data matrix.



Several tessellated data structures exist, however only two are commonly used in GIS's. The most popular cell struc-



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ture is the regularly spaced matrix or raster structure. This data structure involves a division of spatial data into regularly spaced cells. Each cell is of the same shape and size. Squares are most commonly utilized.

Since geographic data is rarely distinguished by regularly spaced shapes, cells must be classified as to the most common attribute for the cell. The problem of determining the proper resolution for a particular data layer can be a concern. If one selects too coarse a cell size then data may be overly generalized. If one selects too fine a cell size then too many cells may be created resulting in a large data volume, slower processing times, and a more cumbersome data set. As well, one can imply accuracy greater than that of the original data capture process and this may



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result in some erroneous results during analysis.

As well, since most data is captured in a vector format, e.g. digitizing, data must be converted to the raster data structure. This is called vector-raster conversion. Most GIS software allows the user to define the raster grid (cell) size for vector-raster conversion. It is imperative that the original scale, e.g. accuracy, of the data be known prior to conversion. The accuracy of the data, often referred to as the resolution, should determine the cell size of the output raster map during conversion.

Most raster based GIS software requires that the raster cell contain only a single discrete value. Accordingly, a data layer, e.g. forest inventory stands, may be broken down into a series of raster maps, each representing an attribute



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type, e.g. a species map, a height map, a density map, etc. These are often referred to as one attribute maps. This is in contrast to most conventional vector data models that maintain data as multiple attribute maps, e.g. forest inventory polygons linked to a database table containing all attributes as columns. This basic distinction of raster data storage provides the foundation for quantitative analysis techniques.

This is often referred to as raster or map algebra. The use of raster data structures allow for sophisticated mathematical modelling processes while vector based systems are often constrained by the capabilities and language of a relational DBMS.

This difference is the major distinguishing factor between



vector and raster based GIS software. It is also important to understand that the selection of a particular data structure can provide advantages during the analysis stage. For example, the vector data model does not handle continuous data, e.g. elevation, very well while the raster data model is more ideally suited for this type of analysis. Accordingly, the raster structure does not handle linear data analysis, e.g. shortest path, very well while vector systems do. It is important for the user to understand that there are certain advantages and disadvantages to each data model.

- **Raster Advantages**

- It is the most common data format
- It is easy to perform mathematical and overlay



operations

- Satellite information is easily incorporated
- Better represents “continuous”- type data

3. Image data

Image data is most often used to represent graphic or pictorial data. The term image inherently reflects a graphic representation, and in the GIS world, differs significantly from raster data. Most often, image data is used to store remotely sensed imagery, e.g. satellite scenes or orthophotos, or ancillary graphics such as photographs, scanned plan documents, etc.

Image data is typically used in GIS systems as background display data (if the image has been rectified and georeferenced); or as a graphic attribute. Remote sensing software makes use of image data for image classification and processing. Typically,

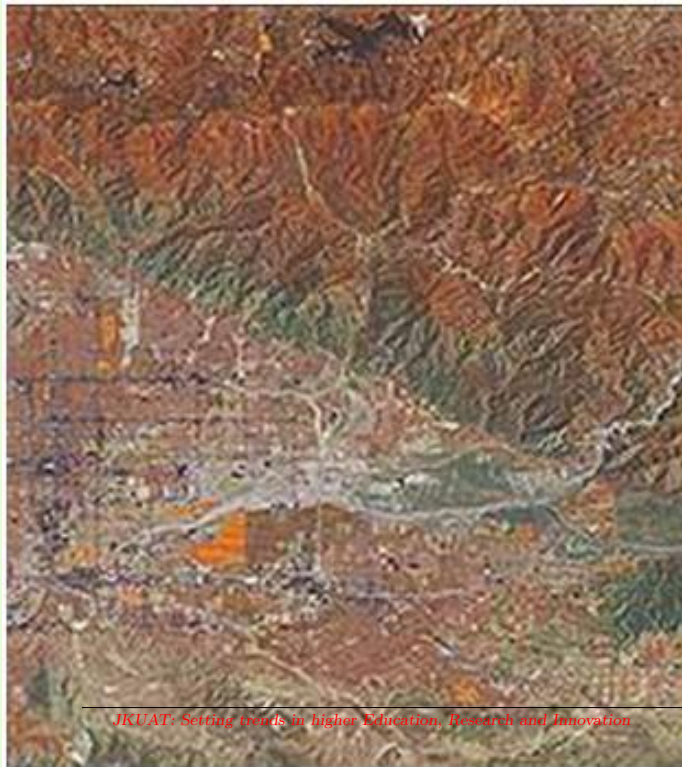


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this data must be converted into a raster format (and perhaps vector) to be used analytically with the GIS. Image data is typically stored in a variety of de facto industry standard proprietary formats. These often reflect the most popular image processing systems. Other graphic image formats, such as TIFF, GIF, PCX, etc., are used to store ancillary image data. Most GIS software will read such formats and allow you to display this data.



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3.4. Attribute data models

A separate data model is used to store and maintain attribute data for GIS software. These data models may exist internally within the GIS software, or may be reflected in external commercial Database Management Software (DBMS). In the raster data model, the cell value (Digital Number) is the attribute. Examples: brightness, landcover code, SST, etc.

For vector data, attribute records are linked to point, line & polygon features. Can store multiple attributes per feature. Vector features are linked to attributes by a unique feature number. A variety of different data models exist for the storage and management of attribute data. The most common are:

- Tabular



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- Hierarchical
- Network
- Relational
- Object Oriented

The tabular model is the manner in which most early GIS software packages stored their attribute data. The next three models are those most commonly implemented in database management systems (DBMS). The object oriented is newer but rapidly gaining in popularity for some applications.



Revision Questions

EXERCISE 1.  Define a map.

Example . Distinguish between vector and raster formats.

Solution: Revise.

EXERCISE 2.  What is a spatial data model?



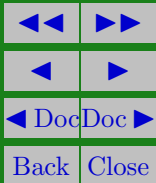
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Solutions to Exercises

Exercise 1. Revise.

Exercise 1





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Exercise 2. Revise.

Exercise 2

